

Summary

This project aims to induce false negatives from Pangram through various AI-implemented adversarial transformations that preserve overall argument and readability. The attacks are organized into three categories, each targeting a different hypothesized detection mechanism: statistical watermark disruption, semantic distribution shift, and stochastic noise injection.

The first method focuses on disrupting token-level statistics that may function similarly to AI watermarking schemes. These variations manipulate lexical choice, regenerated phrasing, and tokenization behavior to alter generation patterns without substantially changing meaning.

The second method targets higher-level semantic organization and discourse structure. Its variations introduce controlled shifts in voice, topical breadth, and conceptual perspective to test whether Pangram relies on a stable, predictable discourse flow commonly associated with AI-generated text.

The third method introduces controlled linguistic variability intended to simulate human-like inconsistency. These variations inject lexical, syntactic, and stylistic noise into essays while preserving readability, testing whether Pangram depends on the relatively smooth and uniform distributions often seen in AI-generated writing.

Together, these methods examine how resilient Pangram is to meaning-preserving perturbations across lexical, semantic, structural, and stylistic dimensions.

Method 1 — Statistical Watermark Disruption

This method is based on the hypothesis that Pangram may rely on token-level statistical signatures similar to AI watermarking schemes, such as the green-red watermark discussed in class. The goal is to perturb the underlying token distributions or tokenization process so that watermark-like patterns disappear. The method evaluates how robust Pangram is to modifications that preserve readability but disrupts statistical fingerprints of AI generation.

- Variation 1 — Lexical Substitution Attack

This variation substitutes synonyms and phrases in the essay but preserves the original meaning and overall structure. Common transition phrases, verbs, and descriptive words are replaced with semantically similar alternatives in order to alter token frequencies.

- Variation 2 — Essay Paraphrasing Attack

This variation rewrites the essays through an LLM's paraphrased regenerations. This approach replaces larger portions of the original token sequence and therefore more aggressively alters the statistical generation distribution underlying the essay.

- Variation 3 — Tokenization Perturbation Attack

This variation introduces Unicode-based or whitespace-based perturbations, such as zero-width characters or homoglyph substitutions, in order to interfere with the detector's tokenization. This attack specifically targets the reconstruction of tokens, testing whether encoding-level manipulations disrupt Pangram's watermark recovery.

Method 1	Baseline	Var 1	Δ Var 1	Var 2	Δ Var 2	Var 3	Δ Var 3
Essay 1	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0
Essay 2	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0
Essay 3	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0

Method 2 — Semantic Drift and Distribution Shift Attack

This method is based on the hypothesis that Pangram may rely on semantic coherence, discourse stability, and predictable information flow, rather than pure token statistics. This method introduces controlled semantic perturbations that preserve the overall topic while shifting tone, meaning distribution, and contextual consistency. The goal is to evaluate how robust Pangram is to semantic transformations that simulate discourse distribution shift.

- Variation 1 — Epistemic Hedging Attack

This variation only introduces qualifiers throughout the essay. Modifiers such as "arguably," "in some sense," "broadly speaking," "to a degree," and "if read this way" are inserted, softening the claims' certainty. This isolates the effect of epistemic complexity for detection, testing whether Pangram relies on the authoritative register that AI-generated academic text tends to exhibit.

- Variation 2 — Topic Expansion Attack

This variation adds analogies, metaphors, and cross-domain comparisons drawn from outside the essay's primary subject. It introduces new conceptual material, such as an analogy to immune system response in an essay about adversarial machine learning. This tests whether Pangram is sensitive to topical breadth and cross-domain semantic mixing, which AI-generated essays may avoid to be sharp on topic.

- Variation 3 — Perspectival Reframing Attack

This variation shifts framing across the essay, alternating among technical explanations, big-picture concepts, and institutional or societal perspectives. Earlier statements are occasionally reinterpreted from a new framing without contradicting — for example, a claim originally presented in technical terms might later be described as a social or organizational issue. This tests whether Pangram sees AI generations as monotonic semantic and discourse progression.

Method 2	Baseline	Var 1	Δ Var 1	Var 2	Δ Var 2	Var 3	Δ Var 3
Essay 1	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0
Essay 2	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0
Essay 3	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0

Method 3 — Stochastic Noise Injection Attack

This method is based on the hypothesis that Pangram may assume AI-generated text to follow relatively smooth, low-noise linguistic distributions. This method introduces controlled stochastic noise into the text while preserving readability and overall meaning. The goal is to evaluate whether Pangram is robust to perturbations that simulate imperfections and variability in human language.

- Variation 1 — Lexical Noise Injection Attack

This variation introduces some random lexical noise into the essay, such as filler or low-information words (e.g., "you know", "I mean", "well", "anyway", "look", "frankly"). The semantic content remains unchanged, but the text becomes more variable at the word level. This tests whether Pangram is sensitive to overly clean lexical distributions that may characterize AI-generated text.

- Variation 2 — Syntactic Noise Injection Attack

This variation creates syntactic irregularities such as parenthetical insertions, mid-sentence digressions, and minor clause restructuring, while maintaining overall sentence length. This targets sentence-level structure, introducing variability in syntactic rhythm.

- Variation 3 — Burstiness and Register Variation Attack

This variation introduces document-level stochastic noise by deliberately varying sentence length distribution and stylistic register. AI-generated text may produce sentences more uniform in length and more consistent in register than humans' does. This variation amplifies that unevenness: some sentences are shortened to fragments or near-fragments, others are extended through additional clauses, and stylistic register is varied between sections (a slightly more formal paragraph followed by a looser one, for example). It directly tests whether Pangram relies on the smoothness of length and register distributions.

Method 3	Baseline	Var 1	Δ Var 1	Var 2	Δ Var 2	Var 3	Δ Var 3
Essay 1	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0
Essay 2	0.0	0.0	+0.0	0.06994	+0.06994	0.0	+0.0
Essay 3	0.0	0.0	+0.0	0.0	+0.0	0.0	+0.0

Analysis

Across 3 baseline essays and 27 attack conditions (3 methods $\times$ 3 variations $\times$ 3 essays), Pangram returned a `factor_human` of 0.0 for all of them except for Method 3 Variation 2 (Syntactic Noise Injection) applied to Essay 2, which sees a delta in `factor_human` of about +0.07.

Strictly based on the numbers, **M3V2 is the most effective while all others tie for least effective. However, it is generous to consider M3V2 effective** because a shift of +0.07 on a 0-1 scale is not sufficient, and Pangram still confidently classifies the essay as AI-generated. Additionally, since the movement occurred on only one of the essays, the improvement might possibly be measurement noise rather than a real signal. Pangram is extremely robust to all three classes of attacks tested.

Method 1 tests whether Pangram depends on token-level statistical signatures corresponding to watermarking schemes. Since word-level substitution, full LLM-based paraphrasing, and character-level Unicode perturbations all produced zero change, **Pangram almost certainly does not try recognizing AI by watermark style of fingerprint**. Specifically, paraphrasing's ineffectiveness suggests Pangram learns something common to LLM-generated texts; since Unicode changes were ineffective, Pangram also **either normalizes inputs or relies on non-tokenization dependent features**.

Method 2 tests whether Pangram depends on semantic coherence, discourse stability, or framing consistency. Since the more modest voices, less focused discussions, and perspective shifts all produced zero change, **Pangram is not primarily reading the discourse-level structure of writing**. Its signal is beyond semantic modification.

Method 3 tests whether Pangram depends on the smoothness characteristic of AI texts. Word-level phatic fillers and document-level amplified burstiness produced no change, but syntactic noise produced marginal movement on one essay. It is **plausible that Pangram has some sensitivity to within-sentence structural regularity**. However, this transformation is also so intricate — modifying the parentheticals, mid-sentence digressions, and clause reorderings — that it may have involved other confounding, uncontrolled-for changes. Additionally, the single +0.07 delta on a single essay is not conclusive.

In conclusion, Pangram is almost certainly impervious to token-level statistics, discourse-level semantics, and surface stylistics. Its training data probably is adversarially augmented, spans across multiple families of LLMs, and stylistically unrestrained on human texts. The most convincing signal it relies on is then the **contextual embeddings from a transformer-based classifier** (as confirmed by their arXiv preprint) that holistically captures AI-like features, patterns of ideas, token sequence fluctuations, and subtle word choices — which are high-dimensional representations that none of my local edits meaningfully disturbed.

References

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